

Introduction to Machine Learning

Success Hacks

- Be prepared for the class by watching the pre class videos
- Attend every class; **Sunday and Thursday**
- Participate in class
- Attempt all the MCQs
- Actively advocate for yourself
- Keep your goal in mind
- **Consistency is the key**

Be patient with yourself !

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Support Features

- **Uplevel** - contains all the learning resources - videos, MCQs, reading materials
- Optional **Post class videos** for additional reference
- **Technical coaching sessions** on Wednesday to clarify all your queries

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Support Features

- **Discord** community for peer learning
- **TAs** to help answer the queries in the live class
- Use **academic support (SSAs)** for any support/advice you want
- **Support tickets** - Raise them any time you want any clarifications/ tech concerns , etc



**INTERVIEW
KICKSTART**

Wednesday and Thursday Sessions

After the main class watch the **post class videos** – These are purely **OPTIONAL**

Wednesday - Technical Coaching Session

One hour of topical review on the class. This is an excellent time to reinforce Sunday's content or ask advanced questions, depending on your experience.

Thursday - Assignment Review Session

Two hours to review the MCQs. MCQs are designed to reinforce the learning objectives which tie directly to knowledge and skills you will need in industry and in interviewing.



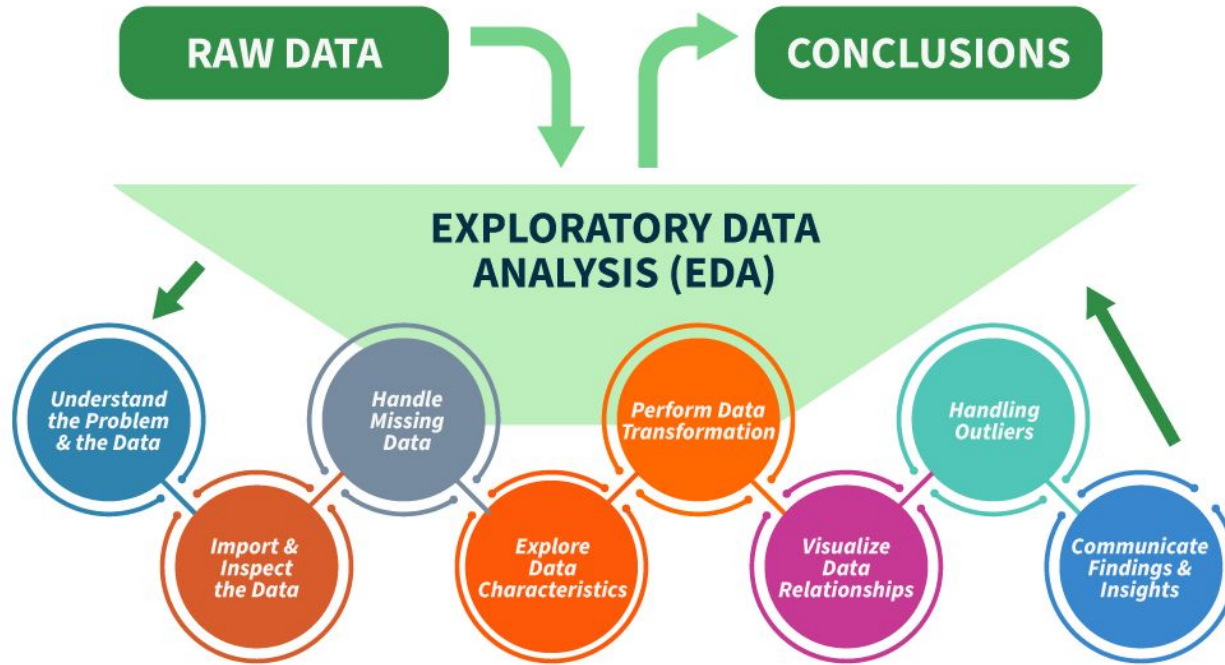
Watched the pre-class foundation videos for this session on **Intro to ML?**

Watch them before the class for better learning outcomes!

Note: The pre-class materials might be videos or reading material or a combination of both. This might vary as per the content of the module.

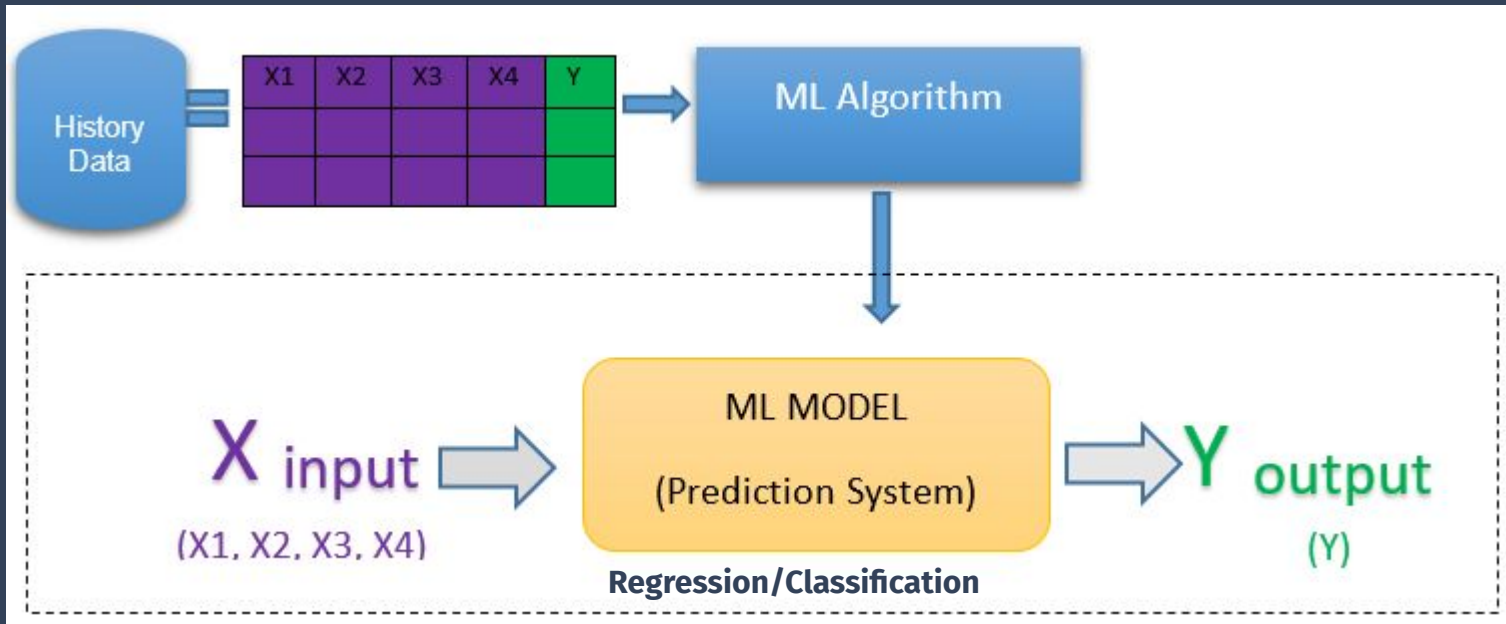
The live class materials also might vary as per the module - slides, coding demo or a combination of both.

What did we learn previously?



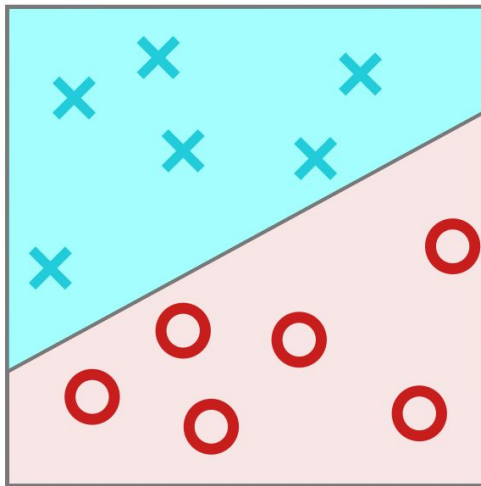
What are we going to learn today?

Model Building



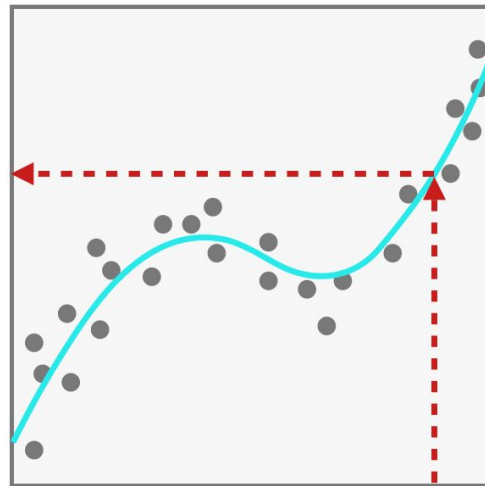
Regression vs Classification

Classification Groups observations into "classes"



Here, the line classifies the observations into X's and O's

Regression predicts a numeric value



Here, the fitted line provides a predicted output, if we give it an input

Linear Regression

What is Linear Regression and why do we need it?

✦ **Predicting numerical values** is a *very widespread problem type*:

we can **solve** these *with ML*!

✦ **Linear regression** is a universal, powerful ML algorithm for such problems

offers *simple, interpretable models* with often *great performance*

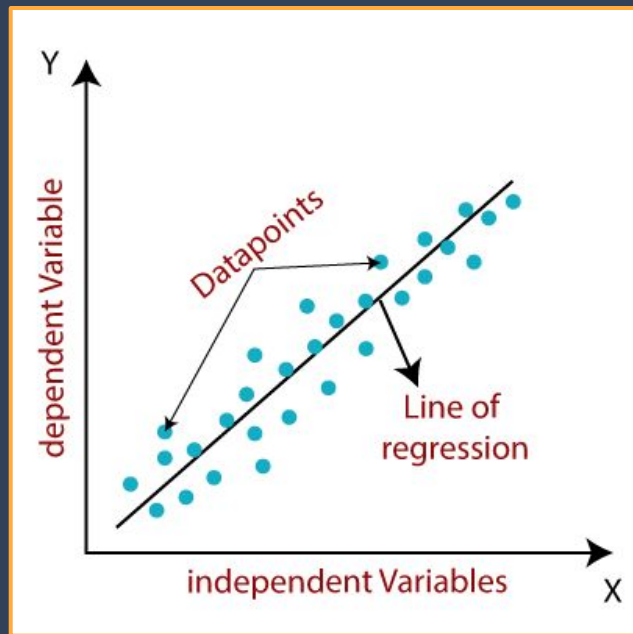
Simple linear regression model

Model the relationship between a **dependent variable** (target) and an **independent variable** (feature)

$$y = \beta_0 + \beta_1 * x_1 + \varepsilon$$

Goal: Find the values of β_0 & β_1 that minimizes the **residual** sum of squares.

Actual value - Predicted value → Errors

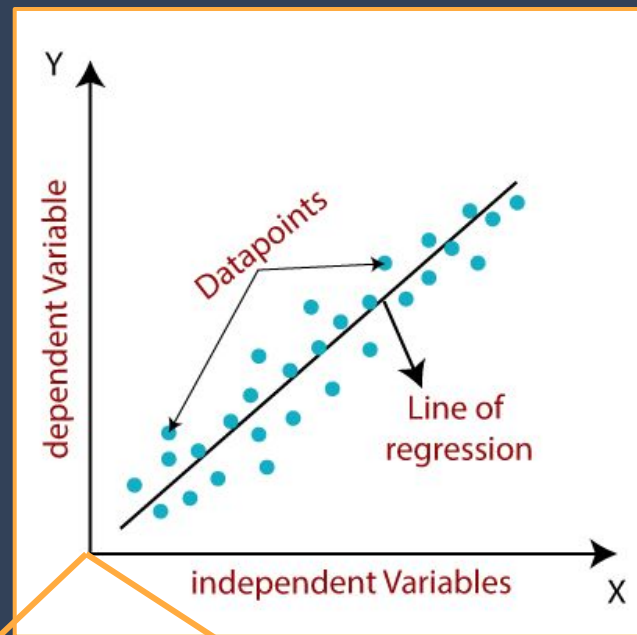


Multiple linear regression model

Model the relationship between a **dependent variable** (target) and one or more **independent variables** (features)

$$y = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2 + \dots + \beta_n * x_n + \epsilon$$

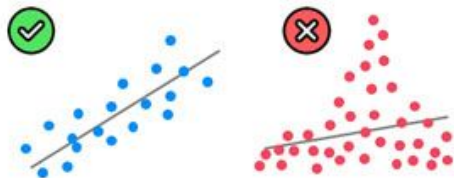
Goal: Find the values of $\beta_0, \beta_1, \dots, \beta_n$ that minimizes the **residual** (actual - predicted) sum of squares.



Linear Regression Assumptions

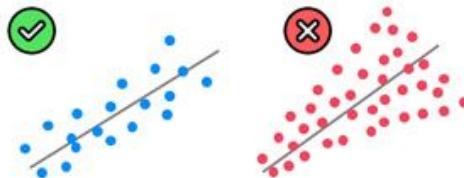
1. Linearity

(Linear relationship between Y and each X)



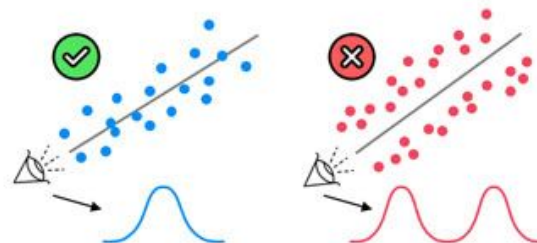
2. Homoscedasticity

(Equal variance)



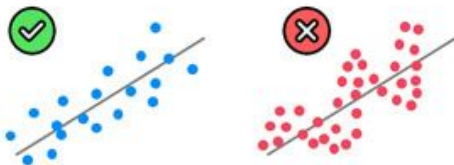
3. Multivariate Normality

(Normality of error distribution)



4. Independence

(of observations. Includes "no autocorrelation")



5. Lack of Multicollinearity

(Predictors are not correlated with each other)



Commonly Used Regression Metrics

📌 Mean Absolute Error (MAE)

📌 Mean Squared Error (MSE)

📌 Root Mean Square Error (RMSE)

📌 R-squared error (R^2)

$y = \text{true value}$ $\dot{y} = \text{predicted value}$ $\bar{y} = \text{average value}$

$$\text{Error} = \text{True Value} - \text{Predicted Value} = y - \dot{y}$$

$$\text{Absolute Error} = |\text{Error}| = |y - \dot{y}|$$

$$\text{Mean Absolute Error (MAE)} = \sum_{i=0}^n \text{Absolute Error} = \frac{1}{n} \sum_{i=0}^n |y_i - \dot{y}_i|$$

$$\text{Mean Squared Error (MSE)} = \frac{1}{n} \sum_{i=0}^n (y - \dot{y})^2$$

$$\text{Root Mean Squared Error (RMSE)} = \sqrt{\sum_{i=0}^n \frac{(y - \dot{y})^2}{n}}$$

$$R^2 = \frac{\sum (\dot{y} - \bar{y})^2}{\sum (y - \bar{y})^2}$$

Logistic Regression

What is Logistic Regression and why do we need it?

✨ **Predicting classes** (that is, classification) is another *very common problem type*:

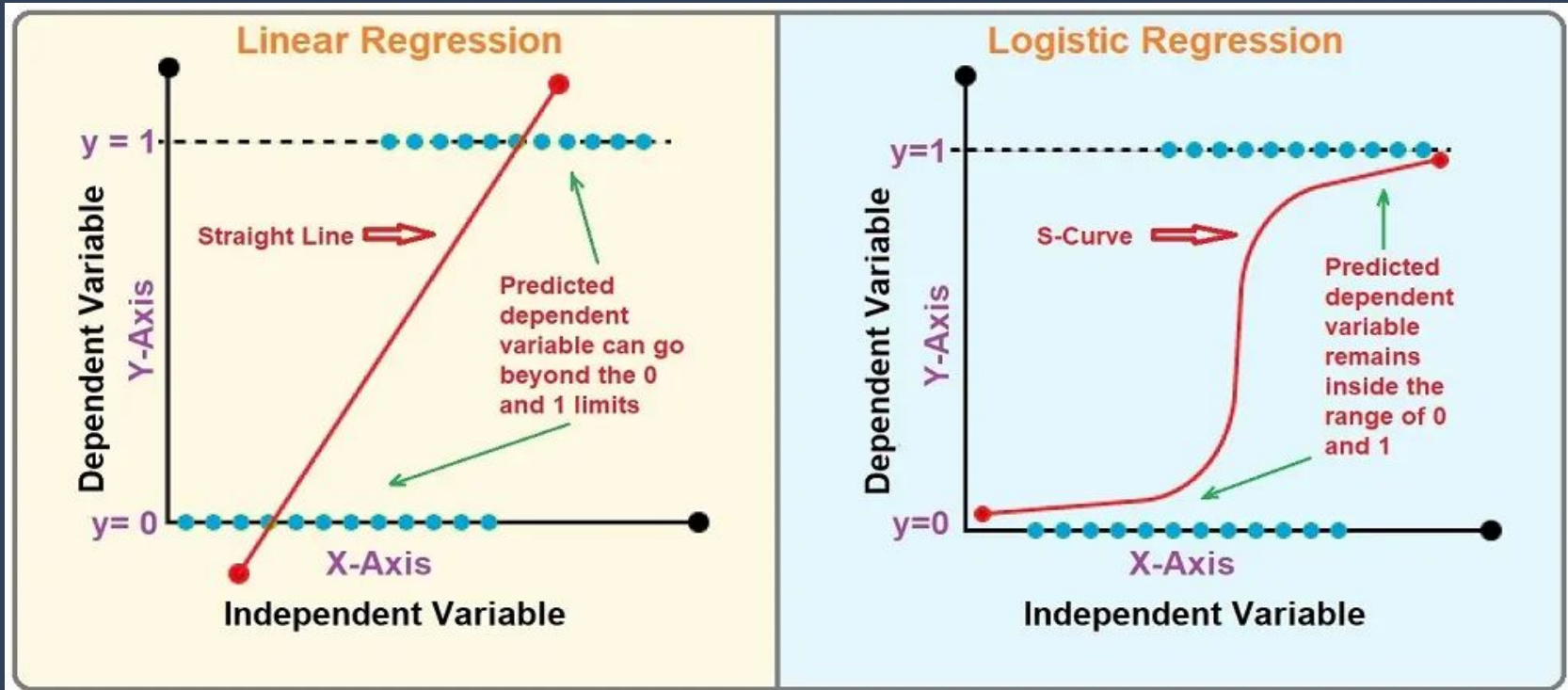
we can **solve** these *with ML!* → *whether something is* **Class A or Class B**

✨ **Logistic regression** is a universal, powerful ML algorithm for such problems

one of the most basic, ubiquitous *models* with often *great performance*

The basis is *binary classification*, easily extendable to *multiclass classification*

Linear vs Logistic Regression



Linear Regression to Logistic Regression

Linear Regression

$$y = b_0 + b_1 * x$$

+

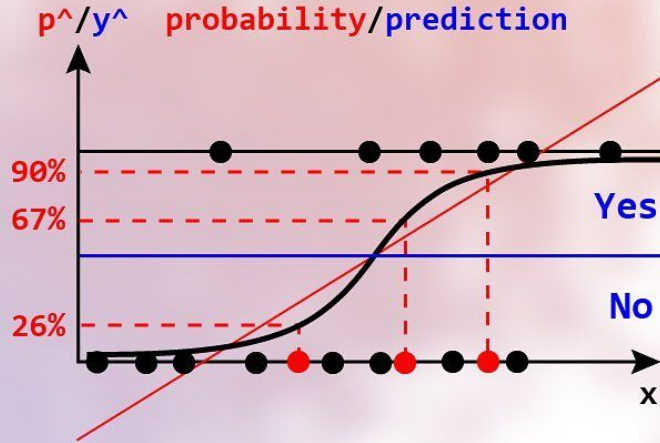
Sigmoid Function

$$p = \frac{1}{1 + e^{-y}}$$

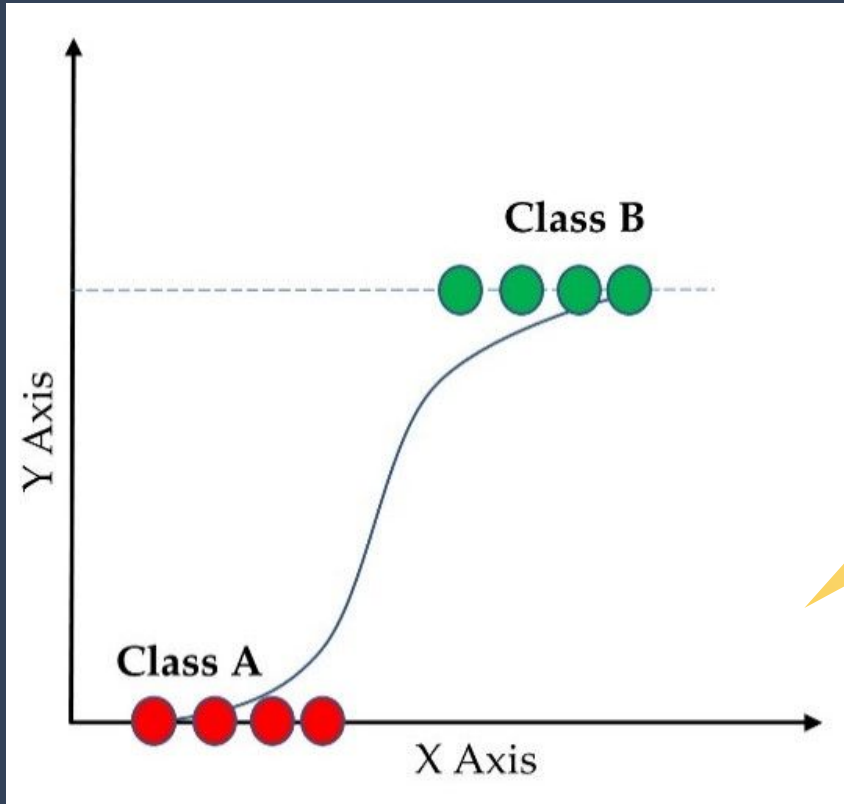
✓

Logistic Regression

$$\ln\left(\frac{p}{1 - p}\right) = b_0 + b_1 * x$$



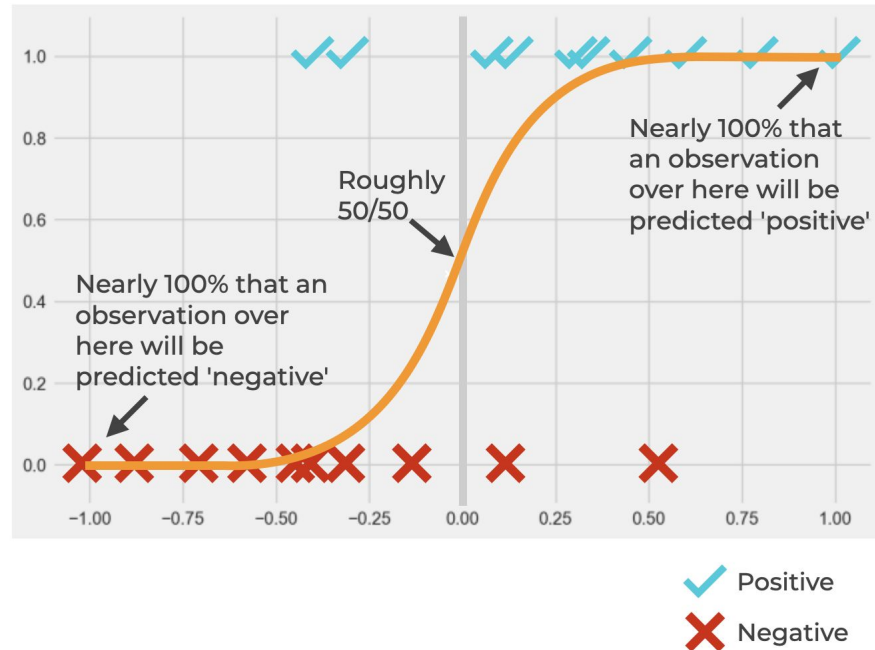
Logistic Regression for Classification



Though the sigmoid function in logistic regression provides a probability value between 0 and 1, **logistic regression is primarily used for classification tasks**, where these probabilities are typically thresholded to make a final classification decision.

Thresholding the values

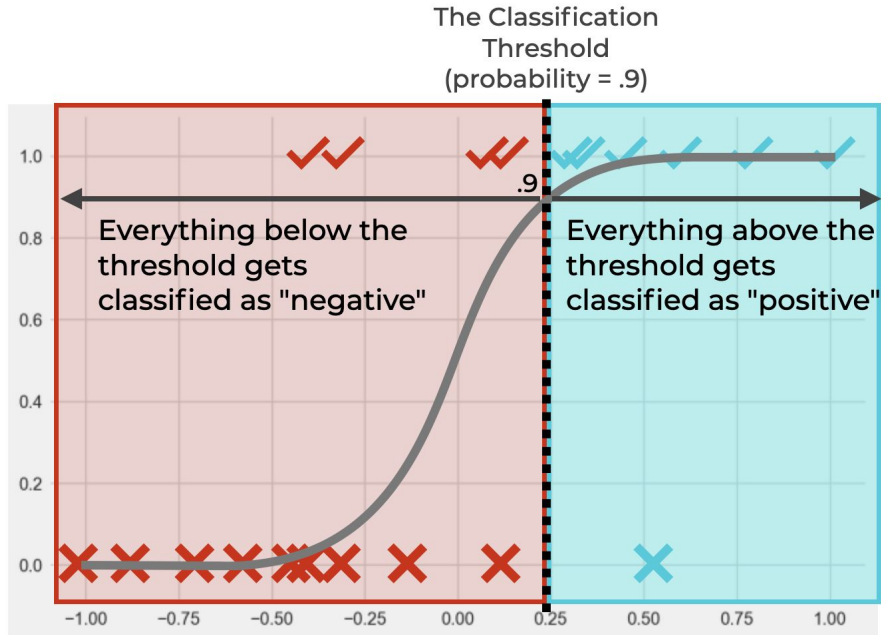
THE VALUE OF THE CURVE REPRESENTS
THE PROBABILITY OF THE "POSITIVE" CLASS



Threshold → decision boundary that determines how the model classifies an incoming example into one of the two groups.

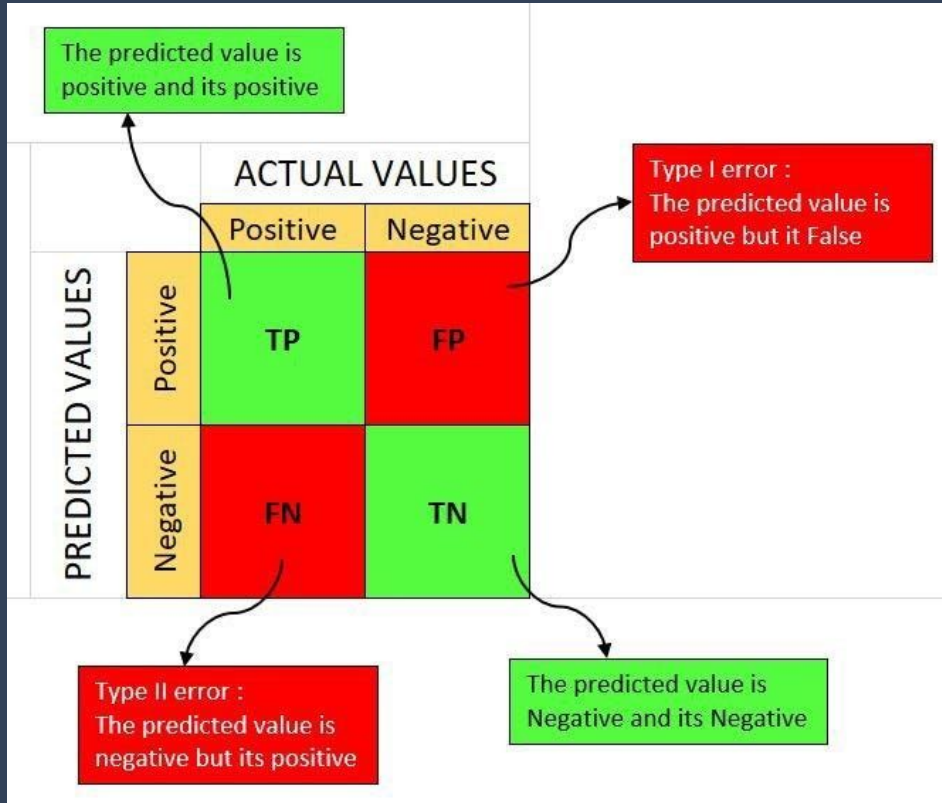
Changing the threshold

CHANGING THE THRESHOLD CHANGES HOW EXAMPLES GET CLASSIFIED



This way of finding the optimal parameters that maximize the likelihood of correctly classifying data points based on the chosen threshold is called as **Maximum Likelihood Estimation**.

Evaluating the model



Confusion Matrix

Classification Metrics

		Actual Value	
		Positive	Negative
Predicted Value	Positive	TP (True Positive)	FP (False Positive)
	Negative	FN (False Negative)	TN (True Negative)

- True Positive (TP) : Observation is positive, and is predicted to be positive.
- False Negative (FN) : Observation is positive, but is predicted negative.
- True Negative (TN) : Observation is negative, and is predicted to be negative.
- False Positive (FP) : Observation is negative, but is predicted positive.

Calculating the metrics

		ACTUAL	
		Positive	Negative
PREDICTED	Positive	True Positive (TP) 5	False Positive (FP) 10
	Negative	False Negative (FN) 15	True Negative (TN) 70

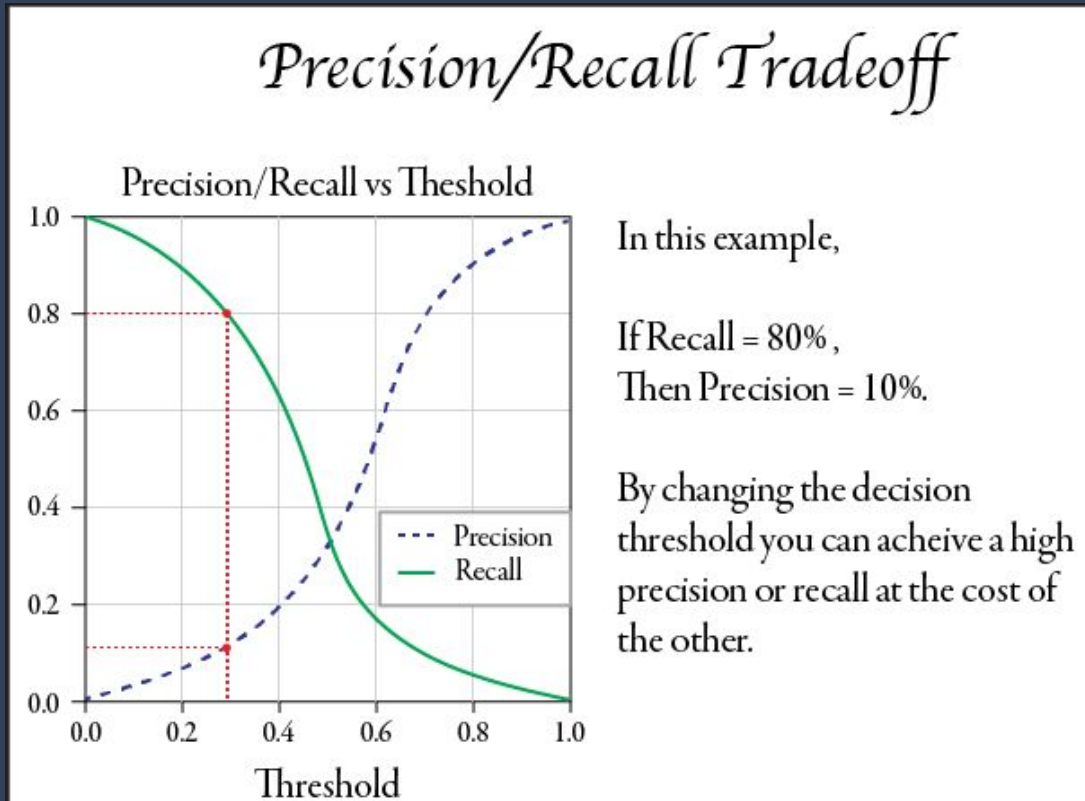
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = 0.75$$

$$\text{Recall} = \frac{TP}{TP + FN} = 0.25$$

$$\text{Precision} = \frac{TP}{TP + FP} = 0.33$$

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = 0.28$$

Precision - Recall tradeoff



Combining Precision & Recall to F1 score



$$F1 = \frac{2 * precision * recall}{precision + recall}$$

$$F1 = \frac{2 \times 0.3 \times 0.1}{0.3 + 0.1} \quad \therefore F1 = 0.15$$

Summary

01

Linear regression predicts a continuous output based on the linear relationship between input variables.

02

Linear regression minimizes the sum of squared errors to fit the best line to the data.

03

Logistic regression predicts a binary outcome by modeling the probability using the sigmoid function & hence performs the classification task.

04

Logistic regression uses maximum likelihood estimation to find the best-fit model.

05

Supervised learning involves training models using labeled data where the input-output pairs are known. Regression & Classification are its types.

Attend the **Thursday sessions** for more practise!

Get your queries clarified via **Wednesday sessions!**

Upcoming Class: **Build Your 1st ML Model**

Note: You will build a text based classification model for performing Sentiment Analysis.

Algorithms Covered: Naive Bayes, K Nearest Neighbours(KNN)

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